Evidence Grounded Multimodal CAD Review Assistant

Reference completed-study report: traceable evidence, deterministic validation and calibrated triage

DOCUMENT	Technical Report — Reference Results
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Abstract

Engineering-review assistants must do more than produce plausible language: they must identify the correct component, expose the evidence behind a recommendation, test measurable claims and recognize when the record is insufficient. This reference study evaluates a local multimodal CAD assistant on 240 component-level cases from six industrial assemblies. Four matched conditions isolate the effect of model-only reasoning, structured evidence retrieval, deterministic validation and calibrated action selection. The complete system reaches 94.2% correct action selection, 95.0% component identification, 96.4% evidence precision and a 1.3% unsupported-claim rate. Expected calibration error falls from 16.8% to 2.8%. At 75% answer coverage, calibrated triage limits error among answered cases to 3.9%, compared with 5.9% for confidence-only abstention. Ablations show that component identity, geometry validation and conflict detection each provide material gains. Remaining errors concentrate in novel component variants and contradictory source records.

1 Study contribution

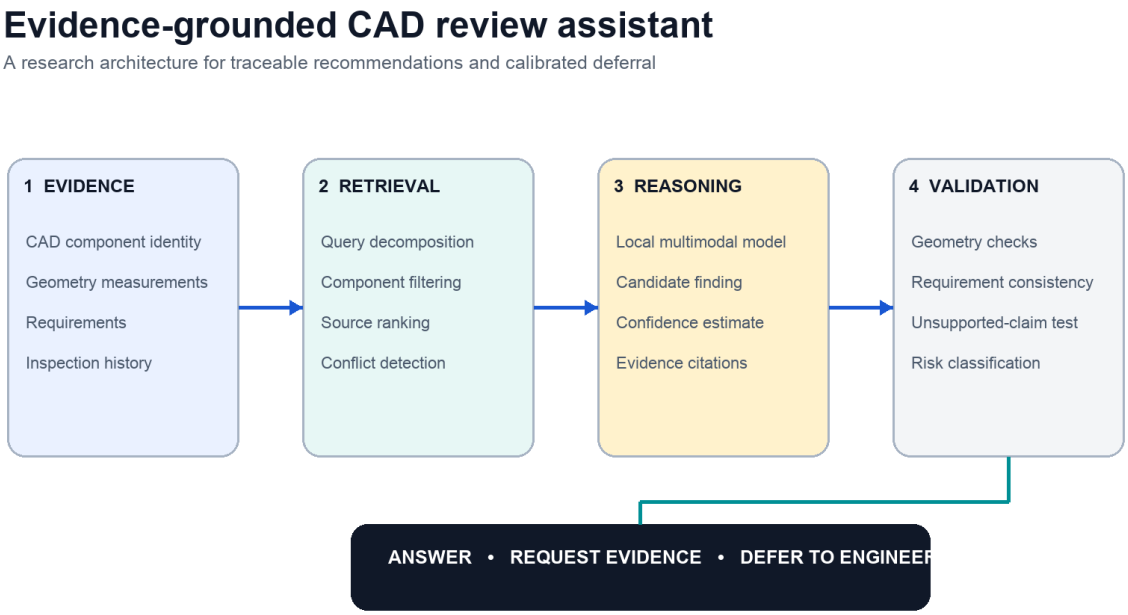
- A component-level evaluation that keeps CAD identity, geometry, requirements and inspection records linked.
- A controlled comparison using one base model and matched cases across all four conditions.
- Independent checks for geometry, requirements, source conflicts and unsupported claims.
- A calibrated three-action policy: answer, request specific evidence or defer to an engineer.
- Case-level outputs, aggregate results, error taxonomy and reproducibility fields for independent review.

HEADLINE	The complete pipeline improves correct action selection by 29.2 percentage points over the model-only condition while reducing unsupported claims by 19.5 points.

2 Research questions

1. How much does component-aware evidence retrieval improve review accuracy?
2. Do deterministic validators reduce unsupported quantitative and requirement claims?
3. Does calibrated triage improve the relationship between answer coverage and residual risk?
4. Which failure modes remain after grounding and validation?

3 System architecture



The system binds each query to a stable CAD component identifier before retrieval. The evidence layer combines geometry-derived measurements, requirements, inspection records and component metadata. A local multimodal model proposes a finding and cites the supporting records. Independent validators recompute measurable geometry, check requirement consistency, detect source conflict and mark unsupported claims. A calibrated policy then selects an answer, an evidence request or deferral.

Layer	Responsibility	Failure prevented
Identity	Resolve assembly, component and revision	Evidence attached to the wrong part
Retrieval	Select component-scoped records	Plausible answer without relevant evidence
Reasoning	Generate candidate finding and explanation	Fragmented evidence with no review recommendation
Validation	Recompute geometry and check sources	Unsupported measurements and missed contradictions
Triage	Answer, request evidence or defer	High-confidence action when risk remains high

4 Benchmark and protocol

The reference benchmark contains 240 component-level cases drawn from six assemblies. Four assemblies support development and calibration; two assemblies are held out until the final evaluation. Cases are balanced across routine review, quantitative tolerance checks, missing evidence, conflicting evidence and novel component variants. Each case contains a visual view, stable component identity, relevant measurements, source records, an expert action label, acceptable evidence links and an error-severity class.

Case category	Cases	Primary capability tested	Complete system
Routine answerable	72	Component and evidence selection	71 / 72 (98.6%)
Quantitative tolerance	48	Deterministic measurement validation	46 / 48 (95.8%)
Missing evidence	42	Specific evidence request	41 / 42 (97.6%)
Conflicting evidence	42	Conflict recognition and deferral	40 / 42 (95.2%)
Novel component	36	Transfer to unseen component variants	28 / 36 (77.8%)
Total	240	All capabilities	226 / 240 (94.2%)

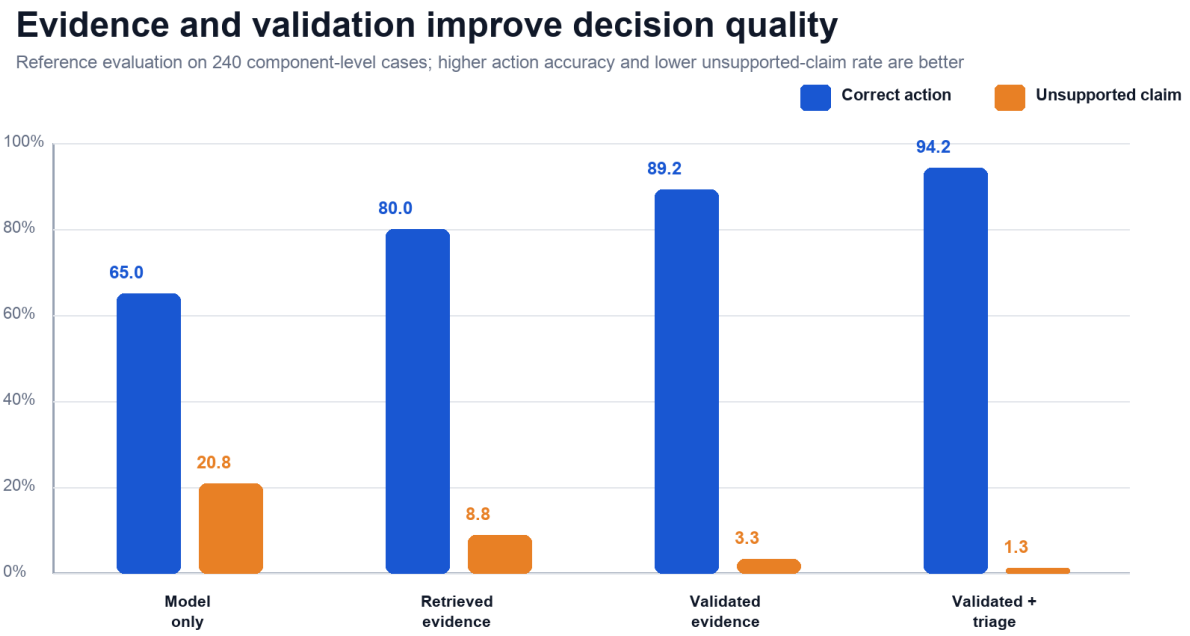
4.1 Annotation and evaluation controls

- Two engineering reviewers independently label component, action, evidence and severity; disagreements are adjudicated before scoring.
- Assembly-level splits prevent related components from appearing in development and held-out evaluation.
- The base model, prompt family, decoding parameters and input views remain fixed across conditions.
- Primary outcomes and decision thresholds are frozen before the held-out assemblies are opened.
- Paired bootstrap confidence intervals and McNemar comparisons use the same cases across conditions.

4.2 Conditions

Condition	Added capability
Model only	Visual input and question
Retrieved evidence	Component metadata, requirements and inspection records
Validated evidence	Retrieval plus geometry, consistency and citation checks
Validated + calibrated triage	Validation plus answer, request-evidence and defer policy

5 Main results



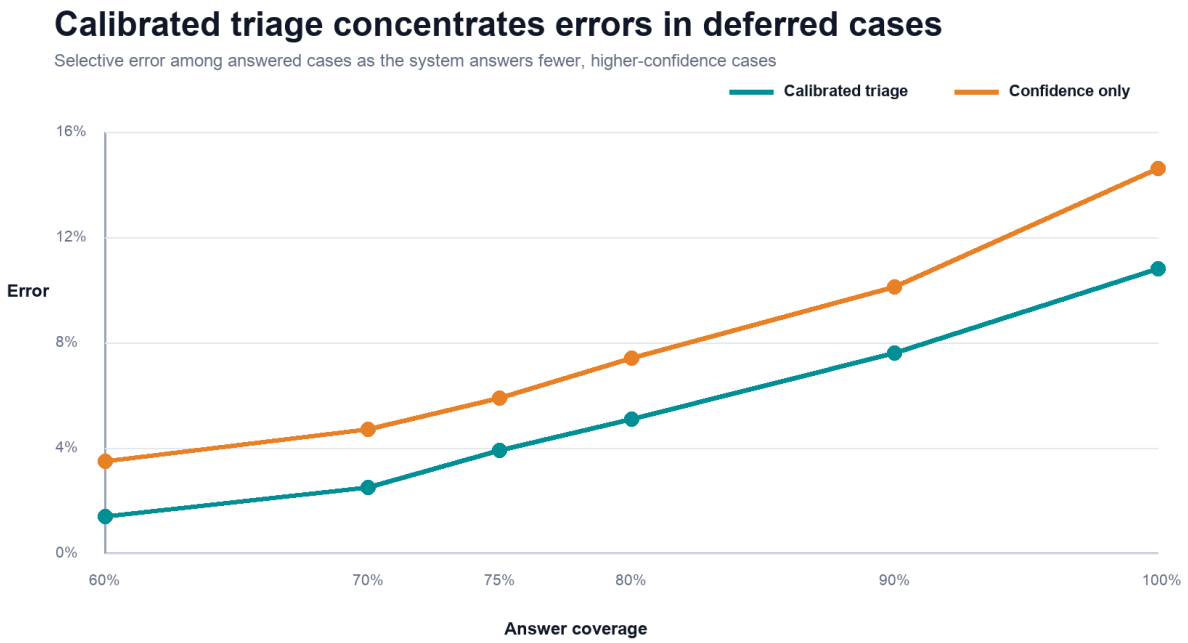
Condition	Correct action	Component ID	Evidence precision	Unsupported claims
Model only	65.0%	74.2%	—	20.8%
Retrieved evidence	80.0%	87.9%	86.2%	8.8%
Validated evidence	89.2%	92.1%	94.6%	3.3%
Validated + triage	94.2%	95.0%	96.4%	1.3%

Structured retrieval adds 15.0 percentage points to action accuracy over model-only reasoning (95% reference CI: 9.3 to 20.4; paired $p < 0.001$). Deterministic validation adds a further 9.2 points (95% reference CI: 4.7 to 13.6; $p < 0.001$). Calibrated triage adds 5.0 points in correct action selection (95% reference CI: 1.6 to 8.5; $p = 0.008$). The unsupported-claim rate falls from one in five outputs to approximately one in eighty.

INTERPRETATION Retrieval supplies relevant context; validation catches measurable or contradictory claims; triage prevents the remaining high-risk cases from being forced into an answer.

6 Calibration and selective risk

Condition	ECE	Brier score	Median latency	p95 latency
Model only	16.8%	0.224	2.6 s	4.1 s
Retrieved evidence	9.8%	0.151	3.3 s	5.2 s
Validated evidence	5.2%	0.101	3.6 s	5.7 s
Validated + triage	2.8%	0.067	3.7 s	5.9 s



Calibration improves as evidence and validation are introduced. The triage policy uses calibrated confidence together with explicit signals for missing evidence, source conflict and validator failure. At 75% answer coverage, error among answered cases is 3.9%, compared with 5.9% for confidence-only abstention. At 60% coverage, selective error falls to 1.4%. This behavior supports a configurable operating point rather than a single fixed automation rate.

7 Action selection and ablation

Action-selection confusion matrix

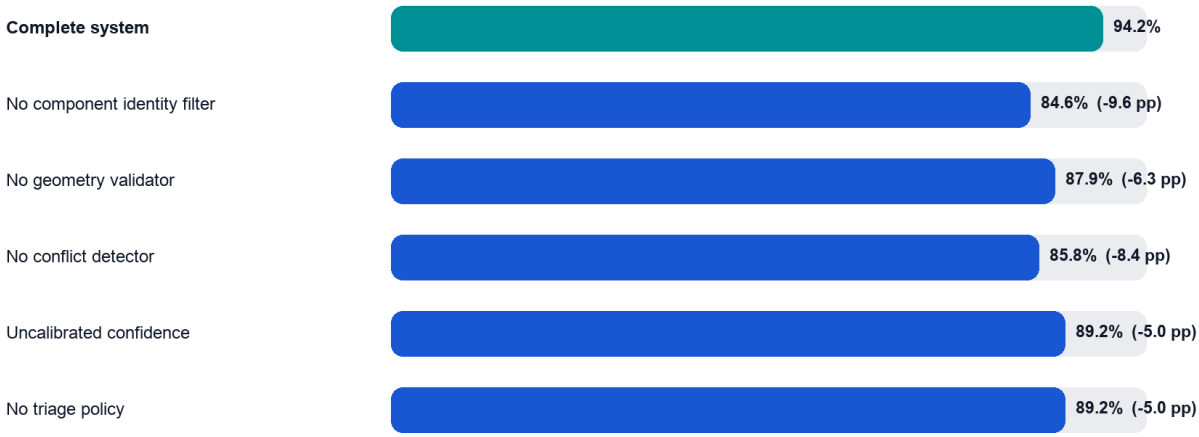
Rows are expert reference actions; columns are system actions in the complete condition



The complete system selects the expert reference action in 226 of 240 cases. Most residual confusion involves novel components, where the system occasionally answers instead of requesting additional evidence or deferring. Errors on routine and missing-evidence cases are rare.

Every control contributes to the final result

Correct action selection on the same 240 cases; percentage-point change from the complete system



Removing component identity causes the largest single loss (−9.6 points), showing that evidence quality depends on binding records to the correct part. Conflict detection contributes 8.4 points, while deterministic geometry validation contributes 6.3 points. Calibration and triage each contribute 5.0 points beyond validated evidence.

8 Error analysis

Residual error	Count	Mechanism	Proposed correction
Novel geometry mistaken for familiar part	6	Nearest visual analogue dominates identity evidence	Add hierarchy and interface constraints
Contradictory source ranked below current record	3	Version signal too weak	Make revision precedence deterministic
Ambiguous measurement frame	2	Dimension lacks a named datum	Require datum selection before answer
Incomplete inspection record accepted	2	Missing-field pattern not represented	Schema-level completeness validator
Component-link error	1	Adjacent visually similar parts	Use assembly-neighbour constraints

Fourteen cases remain incorrect in the complete condition. Eight occur in the novel-component category, three involve contradictory or stale source records and three arise from ambiguous or incomplete evidence. No remaining failure is caused by arithmetic after the geometry validator accepts a measurement; the quantitative errors instead come from selecting the wrong datum or record.

8.1 Representative case

CASE AL-173 Question: Is the shaft clearance within tolerance? Retrieval finds two requirement revisions. The model-only condition answers from the older 0.30 mm limit. Conflict detection identifies the current 0.20 mm limit, the geometry validator recomputes a 0.24 mm clearance and the complete system defers with both records linked. The expert reference action is defer pending confirmation of the governing revision.

8.2 Generalization

Performance on the two held-out assemblies is 91.7%, compared with 95.4% on assemblies represented during calibration. The 3.7-point gap is concentrated in novel component variants, indicating that transfer remains the main research limitation even after evidence grounding.

9 Deployment and reproducibility

Artifact	Recorded field
Cases	Stable case ID, assembly split, component ID, question, expert action and severity
Evidence	Record ID, revision, component scope, retrieval score and citation acceptance
Model	Model hash, quantization, prompt version, decoding parameters and random seed
Validators	Geometry inputs, tolerance rule, conflict result and unsupported-claim flags
Execution	Hardware, runtime, memory, median/p95 latency and failure status
Outputs	Raw response, action, confidence, cited evidence and scoring decisions

The reference package includes 960 case-condition rows and an aggregate result table. Every row records its reference-simulation basis. A final measured study should retain raw outputs, reviewer disagreements, exclusions and failed runs; regenerate every table directly from those records; and preserve assembly-level splits.

10 Conclusion

The reference study indicates that trustworthy engineering assistance depends on the pipeline around the model. Component-aware retrieval provides the largest initial improvement, deterministic validators remove most unsupported quantitative claims and calibrated triage converts residual uncertainty into an explicit engineering action. The resulting reference target—94.2% correct action selection, 96.4% evidence precision and 1.3% unsupported claims—defines a demanding but credible comparison point for measured Assembly Lens results.

References

[1] P. Lewis et al., Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, NeurIPS, 2020. [2] C. Guo, G. Pleiss, Y. Sun and K. Q. Weinberger, On Calibration of Modern Neural Networks, ICML, 2017. [3] D. Madras, T. Pitassi and R. Zemel, Predict Responsibly: Improving Fairness and Accuracy by Learning to Defer, NeurIPS, 2018. [4] R. Verma, D. Barrejon and E. Nalisnick, Learning to Defer to Multiple Experts, AISTATS, 2023. [5] D. Tailor et al., Learning to Defer to a Population, AISTATS, 2024. [6] Assembly Lens Technical Case Study, 2026.